

Unit II

Electromagnetic Theory

Contents

- Del Operator, Gradient, Divergence and Curl
- Gauss's theorem & Stokes theorem (no proof)
- Gauss' law, Ampere's Law, Faraday's Law
- Modified Ampere's Law, Displacement Current,
- Maxwell's equations & their Significance
- Poynting vector $\vec{S}$
- Poynting Theorem (statement & derivation) & energy dissipation
- E.M.W propagation in free space and dielectric medium

Del operator represented by symbol $\vec{\nabla}$ and expressed as

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \quad \text{in Cartesian coordinates}$$

It is a vector differential operator and not a vector quantity.

It can operate on any function in three different ways and the result depends on the manner in which it operates upon the function.

$\vec{\nabla} \phi$	Gradient of a scalar function
$\vec{\nabla} \cdot \vec{F}$	Divergence of a vector function
$\vec{\nabla} \times \vec{F}$	Curl of a vector function

All represent the first derivatives

Gradient of a scalar function $\vec{\nabla}\phi$

When $\vec{\nabla}$ operator acts directly upon a scalar function ϕ , the result is a vector function called Gradient of a scalar function.

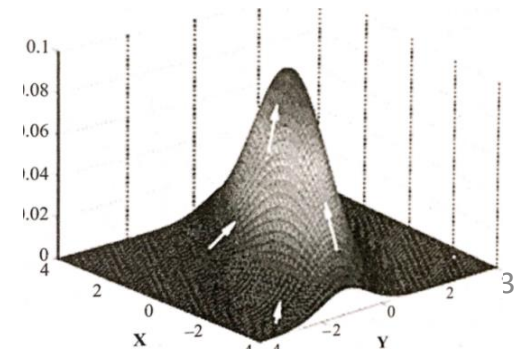
$$\vec{\nabla} \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

It represents a differential of a scalar function ϕ in 3D space.

Geometrical Significance

It has both magnitude & direction. At the point of gradient, its magnitude is the slope along the direction of maximum change of the function with distance & it points along the direction of this maximum rate of change.

Physical Significance It tells how fast the function varies with distance.



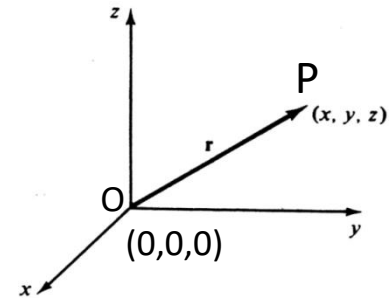
Laminar or lamellar field

If any vector field $\vec{A}$ is expressed as gradient of a scalar field, then $\vec{A}$ is known as Laminar field e.g. Electrostatic field $\vec{E}$ is a lamellar field

$$\vec{E} = -\vec{\nabla}V \quad \text{as it is derived from a scalar potential function } V$$

Q: Find the gradient of 'r' or show that $\vec{\nabla}r = \hat{r}$

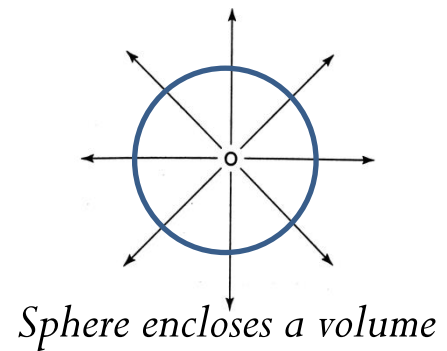
Where 'r' is a distance from the origin O i.e. $r = \sqrt{x^2 + y^2 + z^2}$



$\vec{\nabla} \cdot \vec{F}$ **Divergence of a vector function**

When $\vec{\nabla}$ operator acts on a vector function $\vec{F}$, via dot product, the result is scalar function called Divergence of a vector function.

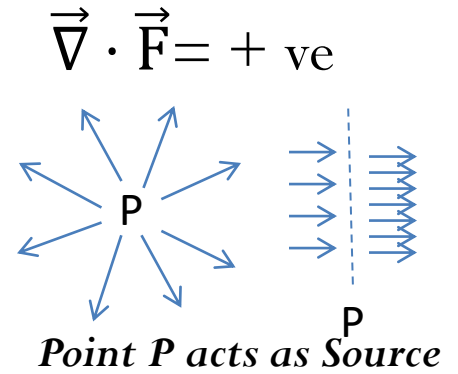
$$\begin{aligned}\vec{\nabla} \cdot \vec{F} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) \\ &= \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \right)\end{aligned}$$



Physical & Geometric Significance At a given point, it measures, how much the vector function $\vec{F}$ spreads out from unit volume over a closed surface.

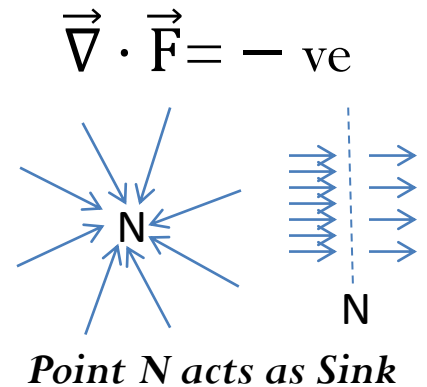
Positive Divergence at a point

If vector field $\vec{F}$ is Spreading out from a point or vector strength increases after passing through a point, then that point acts as a “Source”.



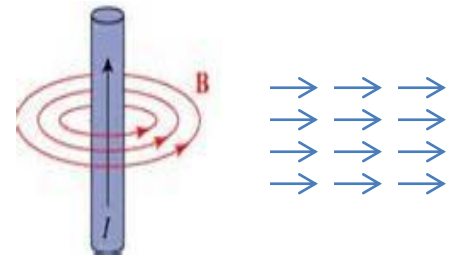
Negative Divergence at point

If vector field $\vec{F}$ is Contracting into a point or vector strength decreases after passing through a point, then that point acts as a “Sink”.



Incompressible fluid or Solenoid field

When the amount of fluid (or flux) entering & leaving is same (there is **no source or sink**) the field is called Incompressible or Solenoid. It represents a field with zero divergence

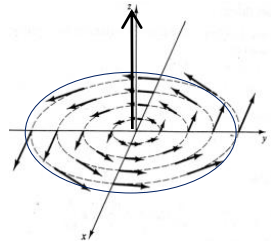


$$\vec{\nabla} \cdot \vec{F} = 0$$

$\vec{\nabla} \times \vec{F}$ **Curl of a vector function**

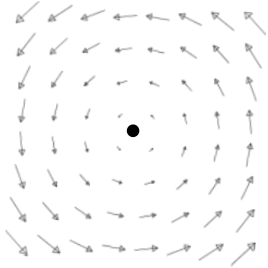
When $\vec{\nabla}$ operator acts on a vector function $\vec{F}$, via cross product, the result is vector function called Curl of a vector function.

$$\begin{aligned}\vec{\nabla} \times \vec{F} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \\ &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \hat{i} - \left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z} \right) \hat{j} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \hat{k}\end{aligned}$$

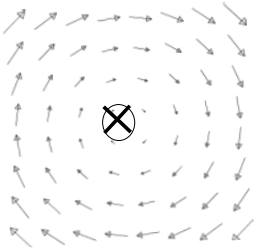


Physical & Geometric Significance It measures how much the vector function $\vec{F}$ is curling around a point in a unit area in 3D space. Its magnitude is equal to the maximum rotation of function $\vec{F}$ /area. It points along the axis of rotation, i.e perpendicular to the area.

The direction of curl vector is given by right hand thumb rule.



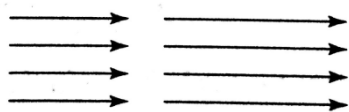
For a vector field rotating anticlockwise,
the Curl vector is pointing out of the page



For a vector field rotating clockwise,
the Curl vector is pointing into of the page

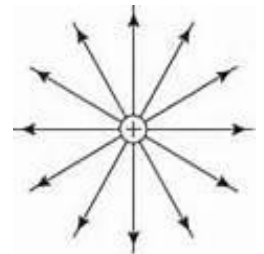


A vector field whose curl is zero is called an Irrotational vector field



$$\vec{\nabla} \times \vec{F} = 0$$

e.g. electrostatic field is an irrotational field $\vec{\nabla} \times \vec{E} = 0$

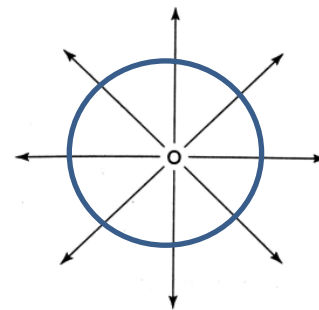


Following theorems are used to transform an expression from volume to surface integral form / vice-versa & surface to line integral form / vice-versa

Gauss's Divergence theorem or Green's theorem or Gauss Transformation formula ($V \rightleftharpoons S$)

It states that the volume integral of the divergence of a vector field $\vec{F}$ over any volume V is equal to the surface integral of the normal component of that vector field $\vec{F}$ taken over a closed surface S , enclosing a volume i.e.

$$\int_V (\vec{\nabla} \cdot \vec{F}) dV = \oint_S \vec{F} \cdot d\vec{a}$$



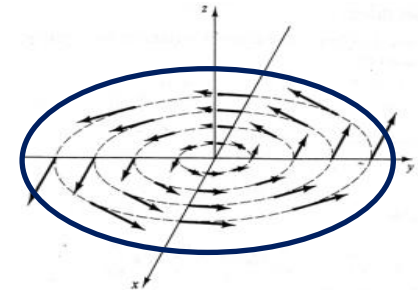
Volume V enclosed
by a Sphere S

Stoke's Law or Stoke's Curl theorem

or Stokes Transformation formula ($S \rightleftharpoons L$)

It states that the surface integral of the curl of a vector field $\vec{F}$ taken over any surface S is equal to the line integral of that vector field $\vec{F}$ taken around the closed curve ℓ , forming the periphery of the surface i.e.

$$\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_{\ell} \vec{F} \cdot d\vec{\ell}$$



Surface area S enclosed
by a closed loop ℓ

Laplacian operator

It is a double differential scalar operator, represented as ∇^2 , given by

$$\nabla^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \quad \text{in Cartesian coordinates}$$

$\nabla^2 \phi$ Laplacian of a scalar is a scalar function

When $\nabla^2 \phi \neq 0$ it is called Poisson's equation

When $\nabla^2 \phi = 0$ it is called Laplace's equation

Laplace's equation is a special case of Poisson's equation

Q Show that curl of the gradient of a scalar field and divergence of the curl of a vector field is always zero, i.e.

$$\vec{\nabla} \times \vec{\nabla} \phi = 0 \text{ \& \; } \vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$$

Fundamental postulates of electricity & magnetism

“The four Maxwell’s equations”

In electrostatic, magnetostatics & electrodynamics, certain experimental results obtained give a connection b/n the electric field ($\vec{E}$) & electric charge (Q) and b/n the magnetic field ($\vec{B}$) & electric current (I) and also the bilateral coupling between electric & magnetic field quantities.

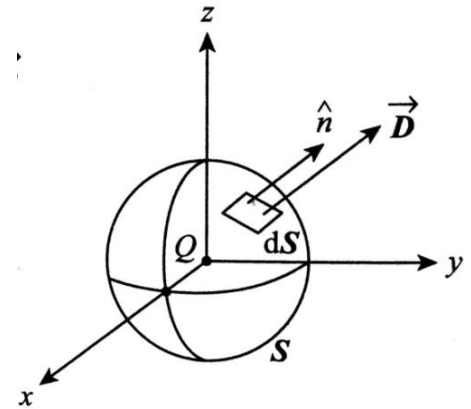
Maxwell encapsulated these connection in the form of mathematical expressions, known as Maxwell’s equations. These equations make the fundamental postulates of electricity & magnetism. These equations can not be derived since these are the generalization of experimental results.

Gauss's law in electrostatics

It gives a relationship b/n the net outward electric flux through a closed surface and the total electric charge enclosed within it.

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon} \quad (1)$$

$$\oint_S \vec{D} \cdot d\vec{a} = q$$



If ρ is volume charge density, then $q = \int_V \rho dV$

The Gauss's law in electrostatic can be written in integral form as

$$\oint_S \vec{D} \cdot d\vec{a} = \int_V \rho dV \quad (2)$$

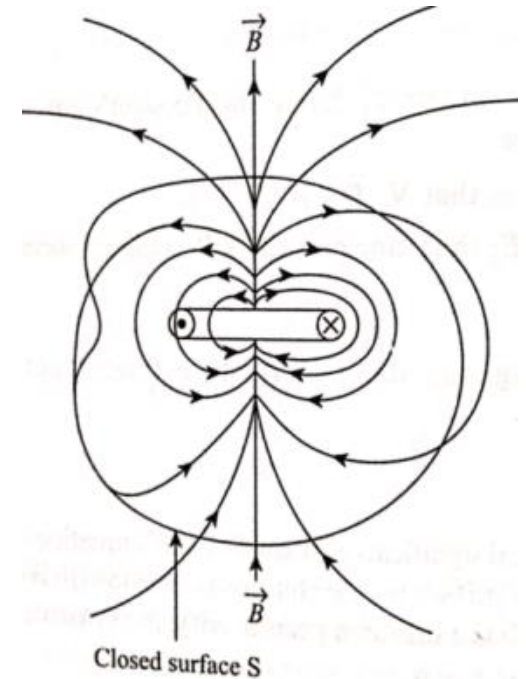
This is the first Maxwell's eq. in integral form.

Gauss's law in magnetostatics

It states that the net outward flux of magnetic induction ($\vec{B}$) through any closed surface is zero.

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

This is the second Maxwell's eq. in integral form. It signifies that isolated magnetic monopoles do not exist, i.e. magnetic field lines form closed continuous loops.

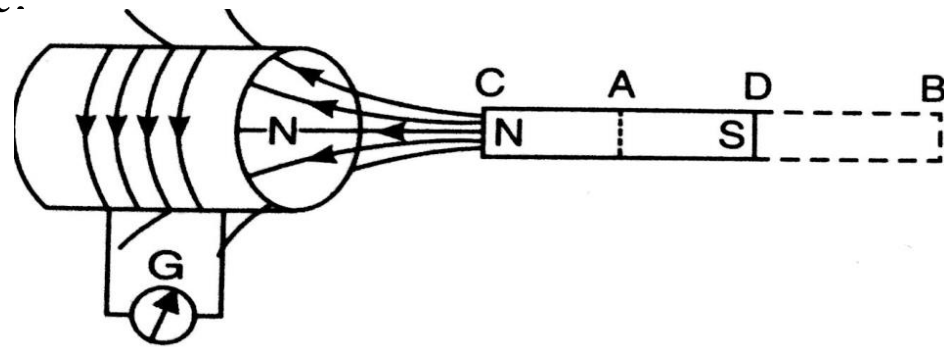


Faraday's law in electrodynamics

According to Faraday's law whenever there is a change in magnetic flux lines through a circuit $\left(\frac{\partial \Phi_E}{\partial t}\right)$, an electro-motive force ($\mathcal{E}$) is induced in the circuit till the change in the flux continues. The magnitude of the induced emf is proportional to the negative rate of change of magnetic flux in the ckt.

$$\mathcal{E} = - \frac{\partial \Phi_B}{\partial t}$$

$$\oint_{\ell} \vec{E} \cdot d\vec{\ell} = - \frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{a}$$



This is the third Maxwell's eq. in integral form

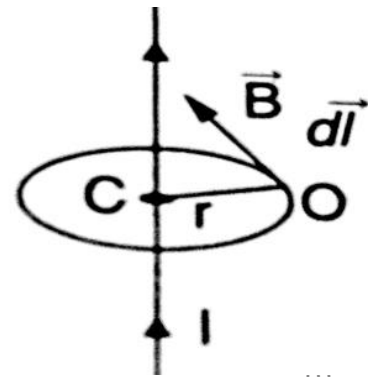
Before discussing 4th Maxwell's equation following topics need to be discussed

- Ampere's law in magnetostatics
- Inconsistency in Ampere's law (observed in 1864)
- Displacement current
- Modified Ampere's law

Ampere's law in magnetostatics

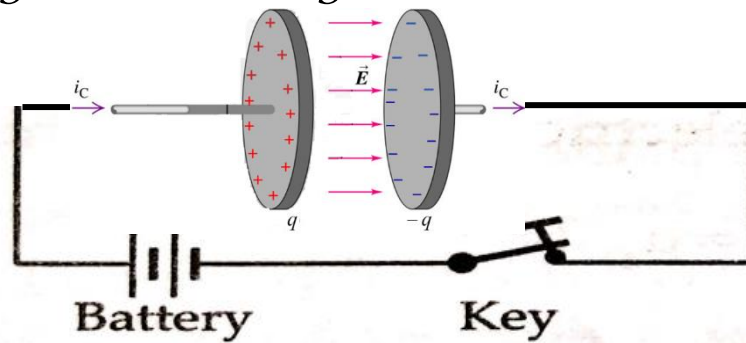
Ampere's law gives a relationship b/n the conduction current flowing in conducting wire and integrated magnetic field, $\vec{B}$ produced around it in any closed loop C.

$$\oint_{\ell} \vec{B} \cdot d\vec{\ell} = \mu I_{\text{enclosed}}$$

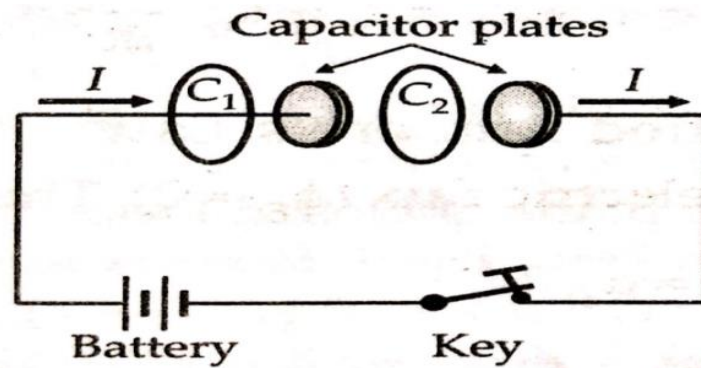


Inconsistency in Ampere's law of magnetostatics

The Ampere's law is valid only for steady state conditions where fields do not change with time. However, the law fails if the field is time varying, such as between the plates of a capacitor, during charging or discharging where the cause of current is changing electric field. *Justification is as follows:*



Consider a parallel plate capacitor being charged by a battery as shown. In the process of charging up of a capacitor, the conduction current I flows in the wire & charge q is supplied on the plates at steady rate, while across the gap there is no actual flow of charges.



To prove the Inconsistency in Ampere's law, Consider two planar loops C_1 & C_2 having same path length ℓ , one around connecting wire & other within the gap b/n the plates respectively. As there is no actual flow of charge across this gap, the conduction current flows through C_1 but not through C_2 .

to loop C_1 $\oint_{C_1} \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enclosed}}$ *Ampere's law applicable*

to loop C_2 $\oint_{C_2} \vec{B} \cdot d\vec{\ell} = 0$ *Ampere's law not applicable*

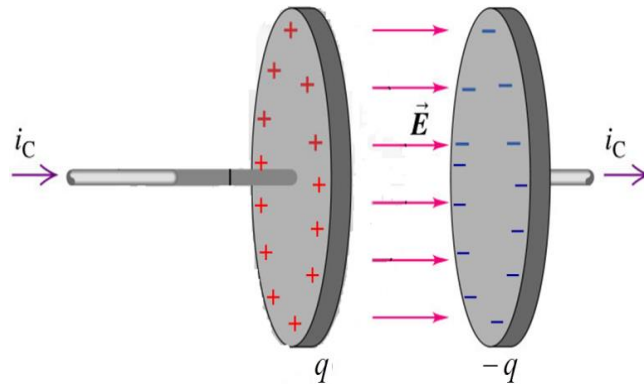
The two contradictory results same circuit go against the continuity of current in the circuit, showing the inconsistency in Ampere's law.

Displacement current

Maxwell realized the need to modify Ampere's law and introduced the concept of Displacement current. Maxwell postulated that during the process of charging/discharging of a capacitor, the changing electric field, within the gap between the plates of a capacitor, is equivalent to a current called the displacement current. Thus the cause of displacement current is the changing electric field.

$$I_d = \epsilon \frac{\partial \vec{E}}{\partial t} A$$

$$I_d = \frac{\partial \vec{D}}{\partial t} A$$



Displacement current density $\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$

It flows in the circuit as long as the electric field is changing.

Modified Ampere's law in electrodynamics:

“It is not only the current in the conductor that produces a magnetic field, but a changing electric field can also produces a magnetic field”. This leads to Modified Ampere's law

$$\oint_{\ell} \vec{B} \cdot d\vec{\ell} = \mu(I_c + I_d)$$

Where I_c is the conduction current flows in a conducting wire, due to the moving charges. While I_d is the displacement current I_d , flowing b/n the plates of a capacitor, the cause of which is the time varying electric field.

$$I_d = \epsilon \frac{\partial \vec{E}}{\partial t} A = \frac{\partial \vec{D}}{\partial t} A$$

The displacement current produces the same magnetic effects as the conduction current, though the cause of their origin is different.

$$\oint_{\ell} \vec{B} \cdot d\vec{\ell} = \mu \left(I_c + \frac{\partial \vec{D}}{\partial t} A \right)$$

if $\vec{J}_c$ represents conduction current density & $\vec{J}_d$ as displacement current density. Then

$$\oint_{\ell} \vec{H} \cdot d\vec{\ell} = \int_s (\vec{J}_c + \vec{J}_d) \cdot d\vec{a} \quad \text{where } \vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

This is fourth Maxwell's eq. in integral form.

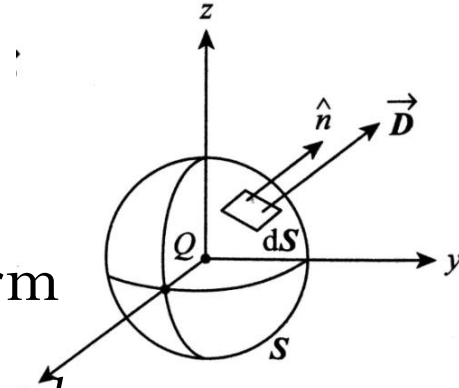
Summary

Maxwell's Equations & their Physical Significance

I. Gauss's law in Electro-statics

$$\oint_S \vec{D} \cdot d\vec{a} = \int_V \rho \, dv$$

Integral form



To transform into a differential form, applying Divergence theorem

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Differential form

It summarizes Gauss's law in Electrostatics, gives a relationship b/n the net outward electric flux through a closed surface and the total electric charge enclosed within it.

II. Gauss's law in Magneto-statics

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

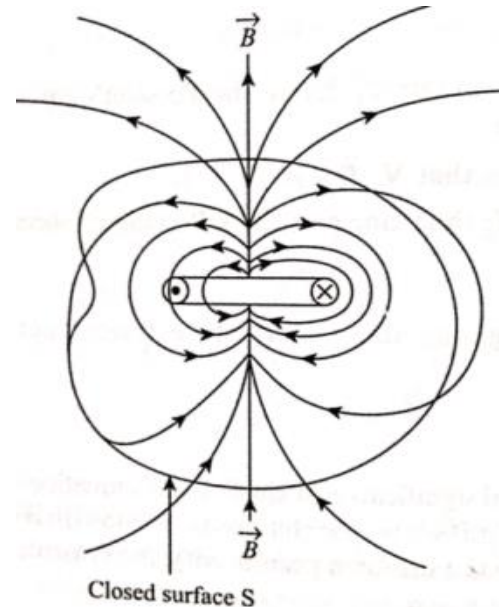
Integral form

To transform into a differential form, applying Divergence theorem

$$\vec{\nabla} \cdot \vec{B} = 0$$

Differential form

It signifies that isolated magnetic monopoles do not exist. There are no sources or sinks for magnetic lines as magnetic field lines form closed continuous loops.



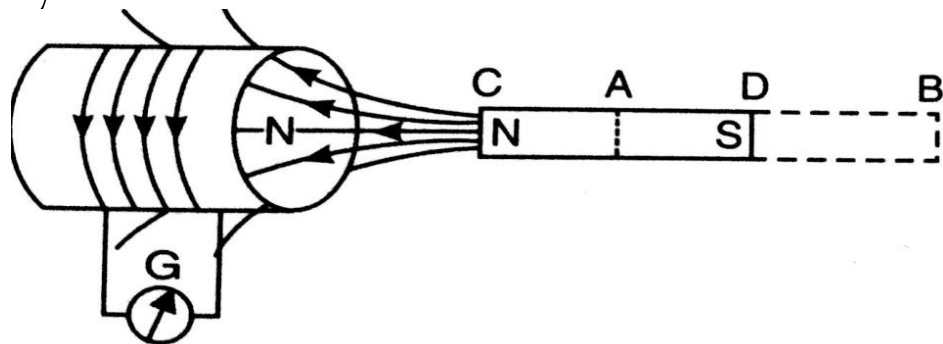
III. Faraday's law of electromagnetic induction

$$\oint_{\ell} \vec{E} \cdot d\vec{\ell} = - \frac{\partial}{\partial t} \int_s \vec{B} \cdot d\vec{a} \quad \text{Integral form}$$

To transform in differential form, applying Stoke's theorem on LHS

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \text{Differential form}$$

It describes Faraday's laws of electromagnetic induction along with Lenz's law. It states that the time varying magnetic field is a source of electric field in space, hence it relates electric field strength $\vec{E}$ with magnetic flux density $\vec{B}$.



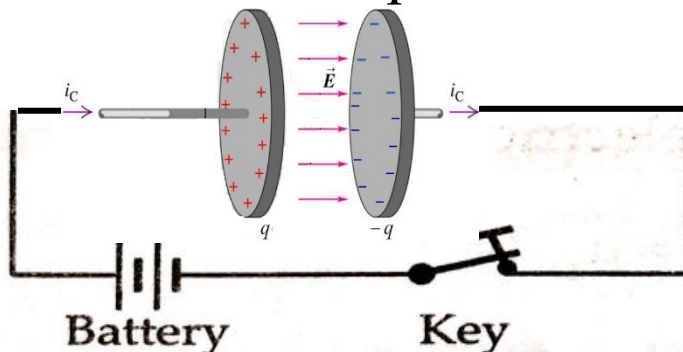
IV. Modified Ampere's law

$$\oint_{\ell} \vec{H} \cdot d\vec{\ell} = \int_s \left(\vec{J}_c + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{a} \quad \text{Integral form}$$

To transform in differential form, applying Stoke's theorem on LHS

$$\vec{\nabla} \times \vec{H} = \vec{J}_c + \frac{\partial \vec{D}}{\partial t} \quad \text{Differential form}$$

It summarizes Ampere's law for time varying currents (modified Ampere's law). It states that “it is not only the current (I) in the conductor that produces a magnetic field, but a time varying electric field can also produces a magnetic field”.

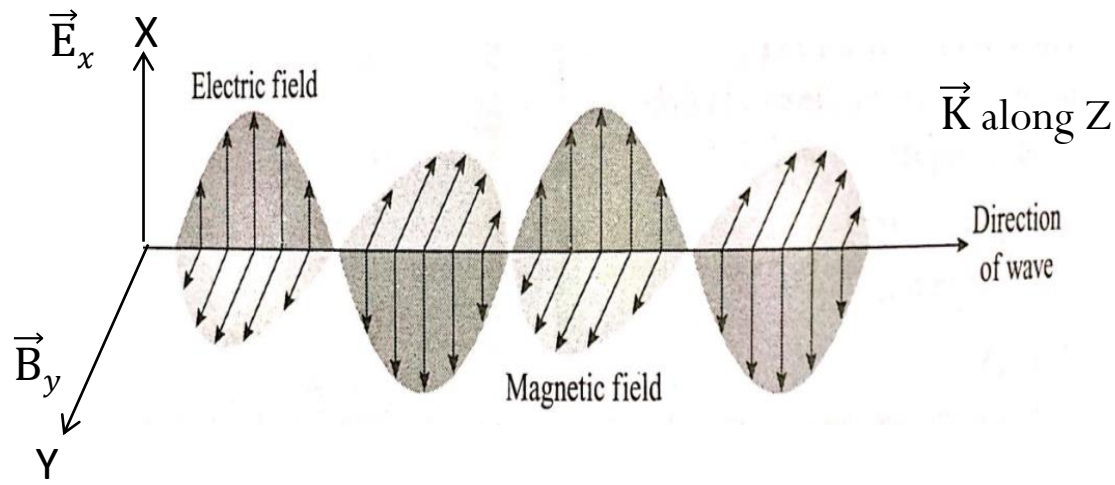


It relates
magnetic field strength $\vec{H}$
with electric flux density $\vec{D}$

Propagation of EM wave

When the charge & current sources varies with time, the electric & magnetic field become interconnected. The bilateral coupling b/n them produces electromagnetic wave which is capable of travelling through free space and material media. The time variation of magnetic field induces an electric field and time variation of electric field in turn induces a magnetic field. Thus electromagnetic fields can exist independently, without electric charges & currents.

The continuous inter-conversion of the fields propagates in space known as electromagnetic waves. The generation of EM-Wave does not require any medium and they propagate through space entirely on their own.



If EM Wave is travelling along z direction, the instantaneous value of Electric & magnetic components are given by

$$\vec{E}_x(\vec{z}, t) = E_0 \cos(\vec{k} \cdot \vec{z} - \omega t) \hat{i} \quad (1)$$

$$\vec{B}_y(\vec{z}, t) = B_0 \cos(\vec{k} \cdot \vec{z} - \omega t) \hat{j} \quad (2)$$

Where E_0 & B_0 represents the maximum Amplitude of Electric & Magnetic field vectors, ω as Angular Frequency & $\vec{k}$, Propagation

vector of wave $\vec{k} = k \hat{k} = \frac{2\pi}{\lambda} \hat{k}$

Poynting Vector $\vec{S}$

Energy emitted/area/time

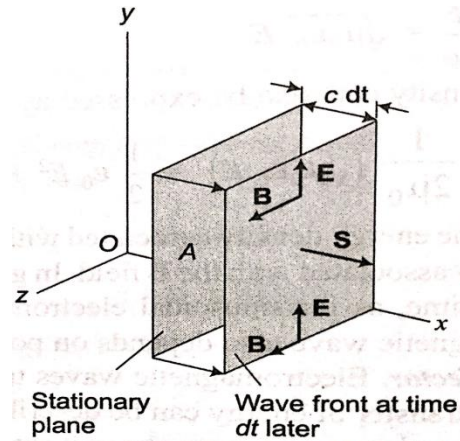
It is defined as the rate at which the energy is transported per unit area by the electric and magnetic field components. It is also known as power density.

Mathematically
$$\vec{S} = (\vec{E} \times \vec{H}) = \frac{1}{\mu} (\vec{E} \times \vec{B}) \quad (\text{watt/m}^2)$$

Where $\vec{E}$ & $\vec{B}$ represents the instantaneous values of Electric & Magnetic field at a point. Its direction is given by cross product rule.

If $\phi_s = \oint \vec{S} \cdot d\vec{a}$ is energy flux, then $\vec{S}$ is Energy flux density

Poynting Theorem talks about “*Law of Conservation of Energy*”



“The total electromagnetic power leaving through the surface is equal to the rate of decrease in the stored energy in the electromagnetic field minus i.e. rate of work done on the charges by electromagnetic forces (i.e. Power dissipation)”.

$$\oint_S \vec{S} \cdot d\vec{a} = - \frac{\partial}{\partial t} \int_V U_{EM} dV - \frac{\partial W}{\partial t} \text{ ----(1)}$$

Rate at which the energy is
flowing out
through the surface

Rate at which the stored energy
in EM-field is decreasing
in a given volume

Rate of work done on
charges by em-forces

Where the term on LHS can be expressed as

$$\oint_S \vec{S} \cdot d\vec{a} = \oint_S (\vec{E} \times \vec{H}) d\vec{a}$$

1st term on RHS as,

$$-\frac{\partial}{\partial t} \int_V U_{EM} dV = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \frac{B^2}{\mu} \right) dV$$

Where Electric energy is the work necessary to assemble static charge distribution against coulomb repulsion of like charges & Magnetic energy, the work required to get current going against back emf

2nd term on RHS as

$$\frac{\partial W}{\partial t} = \frac{\partial}{\partial t} \int_V U_{mech} dV = \int_V (\vec{E} \cdot \vec{J}) dV$$

Since power delivered to the charges will affect their mechanical energy, (Kinetic or potential energy)

As on rearranging it can be expressed as

$$\frac{\partial W}{\partial t} = -\frac{\partial}{\partial t} \int_V U_{EM} dV - \oint_S \vec{S} \cdot d\vec{a} \quad \text{-----(2)}$$

$\therefore$ Poynting Theorem is also known as the “Work-Energy Theorem”.

Derivation of Poynting Theorem:

Suppose a charge & current configuration at time t , produces electric field $\vec{E}$ and magnetic field $\vec{B}$, respectively in a given region.

Suppose dq charges move about a distance $d\vec{\ell}$, in dt time. The amount of work done dW by electromagnetic forces on dq charges in moving a distance $d\vec{\ell}$ is given by $\vec{F} \cdot d\vec{\ell}$

$$dW = (\vec{F}_E + \vec{F}_B) \cdot d\vec{\ell}$$

$$dW = dq(\vec{E} + \vec{v} \times \vec{B}) \cdot \vec{v} dt$$

$$dW = dq(\vec{E} \cdot \vec{v}) dt$$

$$dW = (\vec{E} \cdot \vec{v} dq) dt$$

If ρ is Charge density $\rho = \frac{dq}{dV}$ &

$\vec{J}$ is Current density $\vec{J} = \rho \vec{v}$

$$\vec{J} = \frac{dq}{dV} \vec{v}$$

then $\vec{J} dV = \vec{v} dq$

Rewriting eq (1)

$$dW = (\vec{E} \cdot \vec{v} dq) dt = (\vec{E} \cdot \vec{J} dV) dt$$

Thus, the rate of work done on all charges in some volume V is

$$\frac{\partial W}{\partial t} = \int_V (\vec{E} \cdot \vec{J}) dV; \quad \text{---- (1)}$$

The term $(\vec{E} \cdot \vec{J})$ represents the rate of work done per unit Volume. it can be expressed in terms of field vectors $\vec{E}$ & $\vec{B}$ alone.

From 4th maxwell's eq. $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$

On rearranging $\vec{J} = \frac{1}{\mu_0} \left[\vec{\nabla} \times \vec{B} - \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t} \right]$

$$\vec{E} \cdot \vec{J} = \frac{1}{\mu_0} \left[\vec{E} \cdot (\vec{\nabla} \times \vec{B}) - \epsilon_0 \mu_0 \left(\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right) \right] \text{ ----(2)}$$

From product rule $\vec{\nabla} \cdot (\vec{E} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{B})$

$$\vec{E} \cdot (\vec{\nabla} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{B})$$

$$\vec{E} \cdot \vec{J} = \frac{1}{\mu_0} \left[\vec{B} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{B}) - \epsilon_0 \mu_0 \left(\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right) \right] \text{ ---(3)}$$

From 3rd Maxwell's eq. $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$$\begin{aligned}
\vec{E} \cdot \vec{J} &= \frac{1}{\mu_0} \left[\vec{B} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) - \vec{\nabla} \cdot (\vec{E} \times \vec{B}) - \epsilon_0 \mu_0 \left(\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right) \right] \\
&= - \left[\epsilon_0 \left(\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right) + \frac{1}{\mu_0} \left(\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \right) \right] - \left[\vec{\nabla} \cdot \frac{1}{\mu_0} (\vec{E} \times \vec{B}) \right] \\
&= - \left[\epsilon_0 \left(\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right) + \frac{1}{\mu_0} \left(\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \right) \right] - [\vec{\nabla} \cdot \vec{S}] \\
&= - \left[\epsilon_0 \left(\frac{1}{2} \frac{\partial}{\partial t} E^2 \right) + \frac{1}{\mu_0} \left(\frac{1}{2} \frac{\partial}{\partial t} B^2 \right) \right] - [\vec{\nabla} \cdot \vec{S}] \\
&= - \frac{\partial}{\partial t} \left[\frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \frac{B^2}{\mu_0} \right] - [\vec{\nabla} \cdot \vec{S}] \\
&= - \frac{\partial}{\partial t} [U_{EM}] - [\vec{\nabla} \cdot \vec{S}]
\end{aligned}$$

$$\int_V (\vec{E} \cdot \vec{J}) dV = - \frac{\partial}{\partial t} \int_V U_{EM} dV - \int_V \vec{\nabla} \cdot \vec{S} dV \quad \text{----(4)}$$

Applying Divergence Theorem on the 2nd term of R.H.S

$$\int_V (\vec{\nabla} \cdot \vec{S}) dV = \oint_S \vec{S} \cdot d\vec{a}$$

$$\int_V (\vec{E} \cdot \vec{j}) dV = -\frac{\partial}{\partial t} \int_V U_{EM} dV - \oint_S \vec{S} \cdot d\vec{a} \quad \text{----(5)}$$

rearranging

$$\oint_S \vec{S} \cdot d\vec{a} = -\frac{\partial}{\partial t} \int_V U_{EM} dV - \int_V (\vec{E} \cdot \vec{j}) dV \quad \text{----(6)}$$

This is the Poynting Theorem in integral form

Since the rate of Work done on all the charges in a given Volume affects their mechanical energy (like K.E or P.E), second term on

RHS can be written as $\int_V (\vec{E} \cdot \vec{j}) dV = \frac{\partial}{\partial t} \int_V U_{mech} dV$

where U_{mech} = Mechanical energy density

$$\oint_S \vec{S} \cdot d\vec{a} = -\frac{\partial}{\partial t} \int_V U_{EM} dV - \frac{\partial}{\partial t} \int_V U_{mech} dV$$

$$\oint_S \vec{S} \cdot d\vec{a} = -\frac{\partial}{\partial t} \int_V U dV$$

Applying Divergence theorem on LHS to obtain its differential form

$$\vec{\nabla} \cdot \vec{S} = -\frac{\partial U}{\partial t} \quad \text{-----(8)}$$

This represents the differential form of Poynting Theorem

Here $U = U_{EB} + U_{mech}$

U_{EB} = EM energy density U_{mech} = Mechanical energy density

Propagation of plane EM-Wave in free space

Maxwell, on the basis of Four Maxwell's eqs., predicted that Light is an electromagnetic wave with field vectors $\vec{E}$ & $\vec{B}$, propagating with the speed, $c=3 \times 10^8$ m/sec, in free space.

Proof:

In free space, $\vec{D}=\epsilon_0\vec{E}$ and $\vec{H}=\vec{B}/\mu_0$. Charge density $\rho = 0$ and current density $\vec{J}_c = 0$. Therefore Maxwell's equations are

$$\vec{\nabla} \cdot \vec{E} = 0 \quad (1)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (2)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (3)$$

$$\vec{\nabla} \times \vec{B} = 0 + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad (4)$$

Maxwell's eqs. (3) & (4) are coupled 1st order partial differential eqs. for electric & magnetic field vectors ($\vec{E}$ & $\vec{B}$). To decouple them, applying Curl on both sides.

In eq.(3)

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

$$\vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

from 1st Maxwell eq. $\vec{\nabla} \cdot \vec{E} = 0$

from 4th Maxwell eq. $\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

$$0 - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad (5)$$

Similarly eq. (4)

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = - \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B}) \quad \text{gives}$$
$$\nabla^2 \vec{B} = \mu_o \epsilon_o \frac{\partial^2 \vec{B}}{\partial t^2} \quad (6)$$

Equations (5 & 6) are decoupled, second order partial differential equations for the electric & magnetic field vectors.

Note: In wave mechanics the classical 3D wave equation is given by

$$\nabla^2 f = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2} \quad (7)$$

The field vectors $\vec{E}$ & $\vec{B}$ in eq. (6 & 7) have a form similar to the wave equation (7). Comparing eq. 5, 6 with 7 shows that the field vectors $\vec{E}$ & $\vec{B}$ propagates as wave in free space with velocity given

by $v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/sec} = c$, which is the speed of light

Thus Light is an electromagnetic wave with field vectors $\vec{E}$ & $\vec{B}$

$$\nabla^2 \vec{E} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{and}$$

$$\nabla^2 \vec{B} = \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$

The solution of above wave equation is an oscillatory wave form

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{n} \quad \&$$

$$\vec{B}(\vec{r}, t) = \vec{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{k} \times \hat{n}$$

Show that EM-Wave is Transverse in nature

➤ Solving 1st Maxwell's eq. $\vec{\nabla} \cdot \vec{E} = 0;$

On LHS of above eq., symbols $\vec{\nabla}$ & $\vec{E}$ are represented as

Where $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$ and $\vec{E}(\vec{r}, t) = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$

Solving for $\vec{\nabla} \cdot \vec{E}$

$$\vec{\nabla} \cdot \vec{E} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (E_x \hat{i} + E_y \hat{j} + E_z \hat{k})$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\partial E_x}{\partial x} \hat{i} + \frac{\partial E_x}{\partial y} \hat{j} + \frac{\partial E_x}{\partial z} \hat{k} \quad \text{---- (1)} \quad \text{Where}$$

$$\vec{E}_x = E_{ox} e^{i(\vec{k} \cdot \vec{r} - \omega t)}; \vec{E}_y = E_{oy} e^{i(\vec{k} \cdot \vec{r} - \omega t)}; \vec{E}_z = E_{oz} e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\vec{k} \cdot \vec{r} = (k_x \hat{i} + k_y \hat{j} + k_z \hat{k}) \cdot (x \hat{i} + y \hat{j} + z \hat{k})$$

$$\vec{k} \cdot \vec{r} = (k_x x + k_y y + k_z z)$$

$$\frac{\partial E_x}{\partial x} = \frac{\partial}{\partial x} \left(E_{ox} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right) = i k_x E_{ox} e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\frac{\partial E_y}{\partial y} = \frac{\partial}{\partial y} \left(E_{oy} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right) = i k_y E_{oy} e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\frac{\partial E_z}{\partial z} = \frac{\partial}{\partial z} \left(E_{oz} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right) = i k_z E_{oz} e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

Thus eq (1) becomes

$$\vec{\nabla} \cdot \vec{E} = \frac{\partial E_x}{\partial x} \hat{i} + \frac{\partial E_y}{\partial y} \hat{j} + \frac{\partial E_z}{\partial z} \hat{k}$$

$$\vec{\nabla} \cdot \vec{E} = i(k_x E_{ox} + k_y E_{oy} + k_z E_{oz}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\vec{\nabla} \cdot \vec{E} = i(k_x \hat{i} + k_y \hat{j} + k_z \hat{k}) \cdot (E_{ox} \hat{i} + E_{oy} \hat{j} + E_{oz} \hat{k}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\vec{\nabla} \cdot \vec{E} = i(k_x \hat{i} + k_y \hat{j} + k_z \hat{k}) \cdot (E_x \hat{i} + E_y \hat{j} + E_z \hat{k})$$

$$\vec{\nabla} \cdot \vec{E} = i \vec{k} \cdot \vec{E}$$

$$\vec{\nabla} \cdot \vec{E} = 0 \text{ gives } \vec{k} \cdot \vec{E} = 0 \Rightarrow \vec{k} \perp \vec{E}$$

Similarly

$$\vec{\nabla} \cdot \vec{B} = 0 \text{ gives } \vec{k} \cdot \vec{B} = 0 \Rightarrow \vec{k} \perp \vec{B}$$

Both the Field vectors $\vec{E}$ & $\vec{B}$ being $\perp$ to the direction of propagation vector $\vec{k}$ proves the transverse nature of EM-Waves.

Show that $\vec{E}$, $\vec{B}$ & $\vec{k}$ are orthogonal vectors in EMW

➤ Solving 3rd Maxwell's eq. $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Solving for LHS i.e.

$$\vec{\nabla} \times \vec{E} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{bmatrix}$$

$$\vec{\nabla} \times \vec{E} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{i} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{j} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{k}$$

where

$$\begin{aligned} \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} &= \frac{\partial}{\partial y} \left(E_{oz} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right) - \frac{\partial}{\partial z} \left(E_{oy} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right) \\ &= ik_y E_{oz} e^{i(\vec{k} \cdot \vec{r} - \omega t)} - ik_z E_{oy} e^{i(\vec{k} \cdot \vec{r} - \omega t)} \\ &= i(k_y E_z - k_z E_y) \end{aligned}$$

similarly

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = i(k_z E_x - k_x E_z)$$

and

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = i(k_x E_y - k_y E_x)$$

$$\vec{\nabla} \times \vec{E} = i[(k_y E_z - k_z E_y)\hat{i} + (k_z E_x - k_x E_z)\hat{j} + (k_y E_x - k_x E_y)\hat{k}]$$

$$\vec{\nabla} \times \vec{E} = i(\vec{k} \times \vec{E}) \quad \text{---- (1)}$$

Solving for R.H.S i.e.

$$\begin{aligned} -\frac{\partial \vec{B}}{\partial t} &= -\frac{\partial}{\partial t} \left[\vec{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right] \\ &= - \left[-i\omega \vec{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right] = i\omega \vec{B}. \quad \text{---- (2)} \end{aligned}$$

by substituting eq. (1) & (2), Maxwell's eq. III can be rewritten as

$$\begin{aligned} \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ i(\vec{k} \times \vec{E}) &= i\omega \vec{B} \\ \vec{k} \times \vec{E} &= \omega \vec{B} \quad \text{---- (3)} \end{aligned}$$

This indicates that $\vec{k}$, $\vec{E}$ & $\vec{B}$ are mutually perpendicular each other

Phase between the field vectors $\vec{E}$ & $\vec{B}$

Solving Maxwell's eq. III $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ gives

$$\vec{k} \times \vec{E} = \omega \vec{B}$$

$$\vec{B} = \frac{1}{c} (\hat{k} \times \vec{E}) \quad \text{gives}$$

The magnitude of $\vec{E}$ & $\vec{B}$ related as $B = \frac{E}{c}$

Since the value of c is constant and real, the field vectors are in phase with each other i.e. they have the same relative magnitude at all points at all times

Note: Similar result can be obtained using IV Maxwell's. Eq

$$\vec{\nabla} \times \vec{B} = 0 + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{which on solving gives}$$

$$\vec{k} \times \vec{B} = -\omega \mu_0 \epsilon_0 \vec{E}$$

Wave impedance Z

It is the ratio of magnitude of electric field strength to the magnitude of magnetic field strength. It has unit of 'Ohm'.

$$Z = \frac{|\vec{E}|}{|\vec{H}|} = \frac{\text{volt/m}}{\text{Amp-turn/m}} = \frac{\text{Volt}}{\text{Ampere}} = \text{Ohm}$$

Wave impedance of the EM wave in free space

$$Z = \frac{E}{H} = \frac{E}{B/\mu_0} = \frac{\mu_0 E}{B} = \frac{\mu_0 E}{E/c} = c\mu_0 = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \mu_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

$$Z = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.85 \times 10^{-12}}} = 376.6 \, \Omega \approx 377 \, \Omega$$

In free space, the EM wave experiences a wave impedance of $377 \, \Omega$

This again shows that the ratio of magnitude of $\vec{E}$ to magnitude of $\vec{H}$ is constant & real, hence the field vectors $\vec{E}$ & $\vec{B}$ are in phase with each other.

Instantaneous value of Electromagnetic Energy density in free space

$$U_{EB} = U_E + U_B$$

$$U_{EB} = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{B^2}{\mu_0} \right) = \frac{1}{2} (\epsilon_0 E^2 + \epsilon_0 E^2) = \epsilon_0 E^2$$

$$U_{EB} = \epsilon_0 E^2$$

$$U_{EB} = \epsilon_0 E_0^2 \cos^2(\vec{k} \cdot \vec{r} - \omega t) \quad \text{-----(2)}$$

Instantaneous value of Poynting Vector in free space

$$\vec{S} = \frac{1}{\mu_0} [\vec{E} \times \vec{B}] = \frac{1}{c\mu_0} [\vec{E} \times (\hat{k} \times \vec{E})]$$

$$= \frac{1}{c\mu_0} [\hat{k}(\vec{E} \cdot \vec{E}) - \vec{E}(\vec{E} \cdot \hat{k})]$$

$$= \frac{1}{c\mu_0} [E^2 \hat{k} - 0] = \frac{c}{c^2 \mu_0} E^2 \hat{k} = c(\epsilon_0 E^2) \hat{k}$$

$$= c(U_{EB}) \hat{k} = c[\epsilon_0 E_0^2 \cos^2(kz - \omega t)] \hat{k} \quad \text{-----(3)}$$

Average value of Poynting Vector in free space

In case of light, the wavelength is so short ($5 \times 10^{-7} \text{m}$) & period so brief (10^{-15}sec) that any macroscopic measurement will include many cycles. So all we want is the average values over a complete cycle. In practice, the time averaged value of $\vec{S}$ is more useful, & is called the **intensity of wave**. It is the average power transported/unit area by electromagnetic wave $I = \langle \vec{S} \rangle$

Average value of Electromagnetic Energy density in free space

$$\langle U_{EB} \rangle = \epsilon_0 E_o^2 \langle \cos^2(kz - \omega t) \rangle$$

Where the average value of $\cos^2(kz - \omega t)$ is $\frac{1}{2}$

$$\langle U_{EB} \rangle = \frac{1}{2} \epsilon_0 E_o^2$$

Average value of Poynting vector in free space

$$\langle \vec{S} \rangle = c \langle U_{EB} \rangle \hat{k}$$

$$\langle \vec{S} \rangle = c \epsilon_0 E_0^2 \langle \cos^2(kz - \omega t) \rangle \hat{k}$$

Where the average value of $\cos^2(kz - \omega t)$ is $\frac{1}{2}$

$$\langle \vec{S} \rangle = c \left(\frac{1}{2} \epsilon_0 E_0^2 \right) \hat{k}$$

$$\langle \vec{S} \rangle = c \epsilon_0 \left(\frac{E_0}{\sqrt{2}} \right)^2 \hat{k}$$

$$\langle \vec{S} \rangle = c \epsilon_0 (E_{rms})^2 \hat{k} \quad \text{Where } E_{rms} = \frac{E_0}{\sqrt{2}}$$

$$E_{rms} = E_0 \sqrt{\frac{1}{T} \int_0^T \cos^2(kz - \omega t) dt} = \frac{E_0}{\sqrt{2}}$$

Propagation of plane EM-Wave in dielectric medium

In a dielectric medium $\vec{D} = \epsilon \vec{E}$ and $\vec{H} = \vec{B} / \mu$

charge density $\rho = 0$ and current density $\vec{J}_c = 0$.

Hence Maxwell's equations in dielectric medium

$$\vec{\nabla} \cdot \vec{E} = 0 \quad (1)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (2)$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad (3)$$

$$\vec{\nabla} \times \vec{B} = 0 + \mu \epsilon \frac{\partial \vec{E}}{\partial t} \quad (4)$$

To decouple them, applying Curl on both sides.

In eq.(3) $\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$

$$\vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

from 1st Maxwell eq. $\vec{\nabla} \cdot \vec{E} = 0$

from 4th Maxwell eq. $\vec{\nabla} \times \vec{B} = \mu \epsilon \frac{\partial \vec{E}}{\partial t}$

$$0 - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} \left(\mu \epsilon \frac{\partial \vec{E}}{\partial t} \right)$$

$$\nabla^2 \vec{E} = \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \quad (5)$$

Similarly applying curl on eq. (4) gives

$$\nabla^2 \vec{B} = \mu \epsilon \frac{\partial^2 \vec{B}}{\partial t^2} \quad (6)$$

In wave mechanics the classical 3D wave equation is given by

$$\nabla^2 f = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2} \quad (7)$$

The field vectors $\vec{E}$ & $\vec{B}$ in eq. (6 & 7) have a form similar to the wave equation (7). Comparing eq. 5, 6 with 7 shows that the field vectors $\vec{E}$ & $\vec{B}$ propagates as wave in dielectric medium with velocity v

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0\epsilon_0}\sqrt{\mu_r\epsilon_r}}, \text{ where } c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

$$v = \frac{c}{\sqrt{\mu_r\epsilon_r}} = \frac{c}{\eta}$$

since $\mu_r > 1$ and $\epsilon_r > 1$, $v < c$ The velocity of EM Wave in a dielectric medium is always less than the velocity in free space

In dielectric medium: $\vec{D} = \epsilon \vec{E}$ & $\vec{H} = \vec{B} / \mu$

- Propagation vector $\vec{k} = \frac{2\pi}{\lambda} \hat{k}$ is Real
- $\vec{E}$ and $\vec{H}$ are in phase: $\vec{B} = \frac{1}{v} (\hat{k} \times \vec{E})$, $B = \frac{E}{v}$ is real, const
- Wave impedance (ohms) is a real quantity;

$$Z = \left| \frac{\vec{E}}{\vec{H}} \right| = \frac{\mu \omega}{k} = \mu v = \sqrt{\frac{\mu_o}{\epsilon_o} \frac{\mu_r}{\epsilon_r}} = 376.6 \sqrt{\frac{\mu_r}{\epsilon_r}} = Z_o \sqrt{\frac{\mu_r}{\epsilon_r}}$$

- Electric & magnetic energy density is shared equally,

$U_E = U_B$ so that electromagnetic energy density

$$U_{EB} = \epsilon E^2$$

- Poynting Vector in free space,

$$\vec{S} = v(\epsilon E^2) \hat{k} = v(U_{EB}) \hat{k}$$